

Forecasting at the RBNZ: charting the path forward

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By Paul Conway, Chief Economist

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Introduction

Kia ora koutou. Thank you for tuning in.

It feels serendipitous to be giving a speech on forecasting two weeks after the United States of America imposed sweeping tariffs on its trading partners that were then partly paused for 90 days.

To reiterate key messages made by the Monetary Policy Committee in our *Monetary Policy Review* last week, higher tariffs and uncertainty about global trade policy mean economic activity globally and in New Zealand will most likely be weaker than expected. At this stage, the likely impacts on inflation in New Zealand are more ambiguous, but the balance of risks has shifted to the downside.

We also noted that the Committee has scope to lower the Official Cash Rate (OCR) further, as appropriate, once the impacts of changes in global trade policies on medium-term inflation pressures in New Zealand become clearer.

Of course, we are working to build a deep understand of the likely impacts of changes in global trade policies on the New Zealand economy. Our May *Monetary Policy Statement* will include lots of analysis and insights on what it all means. We look forward to bringing that to you on the 28th of May.

Today I want to speak about how we forecast as part of our monetary policy decision making process at Te Pūtea Matua – the Reserve Bank of New Zealand. Because knowing the current state of the economy and where it is likely headed are key to making interest rate decisions today that keep inflation low and stable in future.

Given the uncertainty we face, forecasting is a critical tool for guiding monetary policy decisions, shaping people's expectations, and ensuring transparency. It is a complex but essential part of monetary policy making.

Our approach to forecasting is continuously evolving as we learn more about the economy and develop new tools through our ongoing research and modelling work. We publish a lot of that work but piecing it together and figuring out what it means for how we forecast can be difficult and I think our approach is not particularly well understood by some.

So, the intention of this speech and supporting material is to improve understanding of why and how we do economic forecasting and how certain aspects of our forecast should be understood.

Along with this speech, we are releasing a Bulletin Article that delves into more detail on how we forecast and two Analytical Notes outlining how we nowcast four important macro variables.¹ We are also launching KiwiGDP, which is a live nowcast of New Zealand's GDP growth in the current quarter. We will update this every week on our website and hope it will be a useful complement to existing models that use innovative techniques to nowcast GDP growth in New Zealand.

¹ Richardson & Williams (2025) 'The Role of Forecasting in Monetary Policy', *Reserve Bank of New Zealand Bulletin*, Vol. 88, No. 1; Bayarmagnai, G. (2025). Nowcasting GDP: A Dynamic Factor Approach. *Reserve Bank of New Zealand Analytical Notes*, AN2025/01; Smith, T. & Lockyer, T. (2025). Focus on the Factor: Applying Dynamic Factor Models in Macroeconomic Forecasting. *Reserve Bank of New Zealand Analytical Notes*, AN2025/02.

I want to focus on three key aspects of forecasting: the importance of forecasting for flexible inflation targeting; how we conduct short-term forecasts (nowcasting); how our forecasts for the OCR should be interpreted.

The key messages I want to convey are:

- We use extensive data and empirical techniques to nowcast the 'starting point' for our economic projections and monetary policy decisions.
- Over the medium-term, our projections rely less on current data and more on our understanding of structural relationships in the economy and how economic events are likely to play out.
- In both cases, we use sophisticated statistical techniques. However, our forecasts are model supported, rather than model driven, with experienced economists playing a vital role in creating our projections.
- Our forecasts for the OCR are based on the full economic picture and are therefore conditional on the economic outlook – as the economic outlook changes, we adjust our OCR forecast such that that inflation still converges to the 2 percent target mid-point.
- And lastly, our forecasts improve transparency, accountability and ultimately the effectiveness of monetary policy.

Forecasting and inflation targeting

To begin at the beginning, the best contribution monetary policy can make to the long-term wellbeing of New Zealanders is to deliver stable prices. The framework we use to deliver price stability is known as flexible inflation targeting. Within this framework, we are tasked with delivering annual CPI inflation between 1-3 percent, with a focus on the 2 percent target midpoint.

The *flexible* bit allows us to deliver price stability while minimising unnecessary disruption to the broader economy and financial system. For example, it would be economically destabilising for us to inflexibly bring inflation back to target within a few months every time we were hit by new economic events. Depending on the shock, being flexible means inflation will not always be at the target midpoint or, in some cases, even within the target range.

Because changes in the OCR affect inflation with long and variable lags, we need to add another word to our framework. That is, we are flexible inflation *forecast* targeters.² There is a forward-looking element to our remit in that we focus on controlling inflation over the medium-term, which we generally regard as being 1-3 years ahead.

It's a bit like football. Chris Wood doesn't waste energy chasing the ball all over the field. Instead, he anticipates where the ball is going to be, rather than where it just was. As a central bank, we need to look ahead and see where the economy is headed and anticipate, rather than look behind us. That's because the economic effects of interest rate decisions made today play out over the next few years. That's how we control inflation. And how Chris Wood scores lots of goals.

² For more on our framework, see rbnz.govt.nz/-/media/project/sites/rbnz/files/monetary-policy/about-monetary-policy/the-monetary-policy-strategy-2024.pdf.

A lot goes in to making this flexible inflation forecast targeting framework work for New Zealanders. We need good research and modelling to understand how the New Zealand economy works and the likely future effects of economic events, including interest rate changes. We need good data and analysis to understand the economic starting point for our projections. And we need good strategies and policy advice to help the Monetary Policy Committee (MPC) make the right decisions.

This framework, which is now used by most central banks globally, was invented right here at the Reserve Bank 35 years ago. We recently held a conference marking the anniversary and exploring the challenges and opportunities for flexible inflation forecast targeting in a post-pandemic world. As I noted in my closing remarks at the conference, the framework remains fit for purpose provided we continue to learn and adapt.³

Does that mean we get it all perfectly right every time? Of course not – our data and analytical tools are imperfect. And it's simply not possible to forecast how black swan events – such as pandemics and trade wars – will exactly play out. But we are back every six weeks or so to update and share our views. Monetary policy is a repeat game, and we learn and adjust as we go.

Nowcasting: understanding the present to predict the future

I want to focus on the work we do to understand the starting point for the economy as part of our forecasting process, which is critical in setting monetary policy. If we don't know where we are now, it's much harder to know where we are going and how to set the OCR consistent with low and stable inflation in future.

Unfortunately, it is not easy to have an accurate or timely view of where the economy stands today. Unlike weather forecasters, who can at least look outside to see if it is raining, economists rely on economic data – such as GDP, employment, and inflation – that arrive with significant lags and are often revised. For example, we receive GDP data about three months after the end of the quarter, by which time the economy has already moved forward.⁴

To address these issues, we use a statistical technique called **nowcasting**. Nowcasting uses a vast array of more timely data – such as from business surveys, financial markets, consumer spending, traffic movements, and internet search trends – to estimate current economic conditions well before the official numbers are released.

Nowcasting allows us to assess economic conditions in near real-time. This is especially important in times of heightened economic uncertainty, such as during financial crises or global shocks, when rapid changes in the economy require a swift and agile monetary policy response.

³ All conference material, including videos of every session, are available at: [35 years of flexible inflation targeting – challenges and opportunities – Reserve Bank of New Zealand – Te Pūtea Matua](#).

⁴ These issues can be particularly troublesome in New Zealand, where the quality of some of our economic data is worse than that of some other OECD economies.

Nowcasting in practice

If I made that sound easy or straightforward, I didn't mean to. Complex statistical issues need to be addressed when building good nowcasting models and frameworks:

- **Data Frequency:** Economic data are reported at different intervals. Some data, like electronic card transactions, are available daily, while others, such as GDP, are only released quarterly. Our nowcasting models integrate these mixed frequencies to construct a coherent picture of current economic activity.
- **Timeliness of Data Releases:** Because different data sets become available to us at different times, our models must continuously update forecasts as new information arrives. This means that a nowcast for a given quarter evolves significantly as more and more data are incorporated.
- **Separating Signal from Noise:** High-frequency data can be volatile due to noise or temporary economic volatility that does not reflect broader trends. For example, a short-lived dip in retail spending due to a weather event is not necessarily indicative of an economic downturn. Our models and expert judgment help filter out this noise.

Importantly, we do not rely on any single model to nowcast the economy. Different models have different strengths, and we combine insights from multiple approaches to produce our best possible estimates. This includes simple indicator regressions to much more advanced econometric techniques that use very large data sets. Some of these nowcasting frameworks have been in use at the Reserve Bank for many years.⁵

As I mentioned earlier, we are publishing the output from one of these models – KiwiGDP – alongside this speech. For the nerds out there, KiwiGDP is a dynamic factor model a lot like one published by the Federal Reserve Bank of New York.⁶ It uses a broad range of economic indicators as they are released to update a nowcast of the unreleased GDP growth rate for the current quarter. We will update KiwiGDP each week, and hopefully it will be a good complement to other existing models using innovative techniques to nowcast GDP growth in New Zealand.⁷

Our nowcasts are **model-supported, not model-driven**. While sophisticated statistical techniques form the backbone of our approach, experienced economists play a vital role in creating our nowcasts. Our economists draw on their knowledge of the models' limitations and a wealth of off-model data, research and industry insights to improve nowcasting results.

Nowcasting vs forecasting

We use nowcasting to help assess economic conditions in the very near term, which is usually the quarter we are in as we compile the projections plus the next quarter. As we look further into the future, we rely less on current data and more on our understanding of structural relationships in the economy and how economic events are likely to play out.

NZSIM – which stands for New Zealand Structural Inflation Model – sits at the core of our medium-term forecasting framework. NZSIM is a semi-structural dynamic stochastic general

⁵ For example, see Richardson, Van Florenstien Mulder and Vehbi (2021), 'Nowcasting GDP using machine-learning algorithms: A real-time assessment', *International Journal of Forecasting*. Karagedikli and Ozbilgin (2019), Mixed in New Zealand: Nowcasting Labour Markets with MIDAS, RBNZ Analytical Note. An2019/04.

⁶ New York Fed Staff Nowcast - FEDERAL RESERVE BANK of NEW YORK.

⁷ KiwiGDP is available at rbnz.govt.nz/kiwi-gdp. Broadly similar nowcasting models are available from Westpac and GDP-Live.

equilibrium model that brings together our research on how the New Zealand economy works, including on the effects of monetary policy on household and firm behaviour.

As with our nowcasts, our medium-term forecasts are model-supported, rather than model-driven. NZSIM provides a rigorous economic foundation, but expert judgment remains essential. In essence, we use NZSIM to structure our insights from staff and MPC into a consistent medium-term outlook for the New Zealand economy.

This blend of quantitative analysis and expert insight ensures our forecasts are timely and robust, giving MPC a clearer, more comprehensive view of the economic outlook. It allows us to explore how economic shocks are likely to evolve over time and guides us on how monetary policy should be set to influence inflation and economic activity going forward.

The OCR forecast is conditional rather than certain

I hope I have made it clear that our interest rate decisions are not based solely on official data, which tend to be lagged and prone to significant revision. We also use a huge wealth of alternative data, combined with empirical techniques and expert judgement, to understand the economy's starting point and likely future path as input into an OCR decision.

I also want to clarify how our forward path for the OCR should be interpreted. A key element of our *flexible inflation forecast targeting* policy strategy is that we publish what we call an 'endogenous' policy interest rate track every three months as part of our projections.⁸ This is our best estimate of how the OCR will need to change over the forecast period to meet our inflation objective, conditional on the economic outlook.

That last bit – conditional on the economic outlook – should be read as being bolded, highlighted, and jumping off the page like a neon sign. I cannot overstate the importance of this conditionality. If the economic outlook changes, which it almost always does to some degree, then our projection for the OCR will also change.

Because the OCR projection is conditional on everything else in the economy evolving as expected, it should almost never be interpreted as a guarantee of future MPC decisions.⁹

The OCR forecast changes with changes in inflation pressure

In our projections, the outlook for the OCR is determined such that the MPC meets its inflation target over the medium term. That is, our inflation forecasts almost always settle close to the 2 percent target midpoint by the end of the projection period. This can be seen in the evolution of Reserve Bank inflation forecasts over time (Figure 1).

If the inflation outlook *did not* always settle at the target midpoint, our forecasts could imply that monetary policy was not being set appropriately to meet our inflation objective. The exception would be if our secondary considerations in setting monetary policy meant that inflation should return to target over a time frame beyond our normal 3-year forecast horizon.

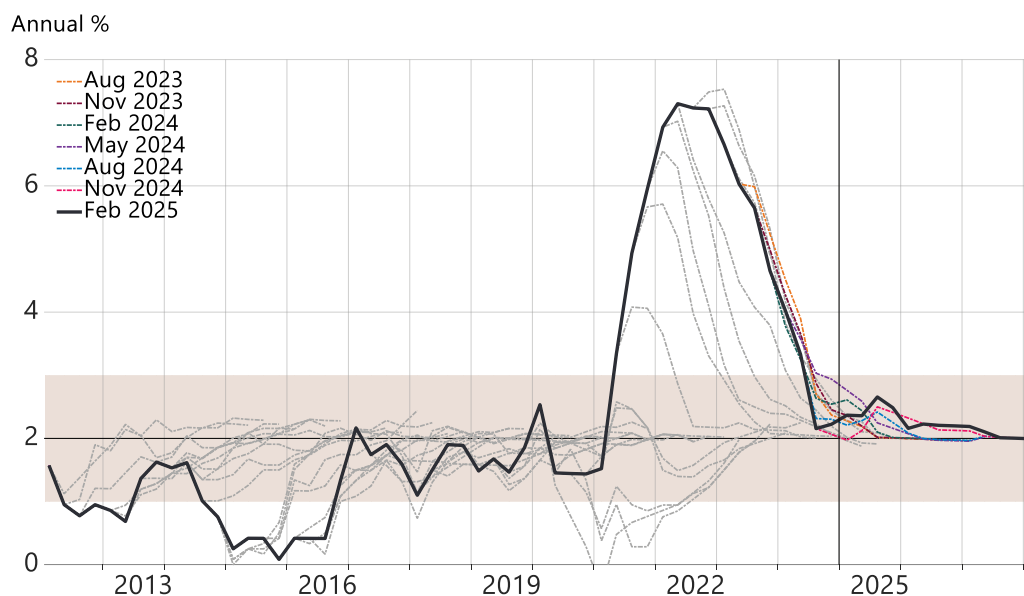
⁸ See rbnz.govt.nz/-/media/project/sites/rbnz/files/monetary-policy/about-monetary-policy/the-monetary-policy-strategy-2024.pdf for more on this strategy and framework.

⁹ An exception to this was during the COVID-19 pandemic in 2020, when the MPC committed to holding the OCR at 0.25 percent for 12 months. Such a commitment reflected the highly unusual and uncertain period the New Zealand and global economies were facing, and the MPC's intent to provide certainty to businesses, households and financial market participants. As discussed in the 2022 *Review of the Formulation and Implementation of Monetary Policy*, such strong commitments —while effective— should be used with caution.

While our forecasts for inflation always converge to 2 percent, our OCR forecast changes significantly through time (Figure 2). Broadly speaking, it is changes in our outlook for the OCR that tell you whether underlying inflationary pressure in the economy has changed.

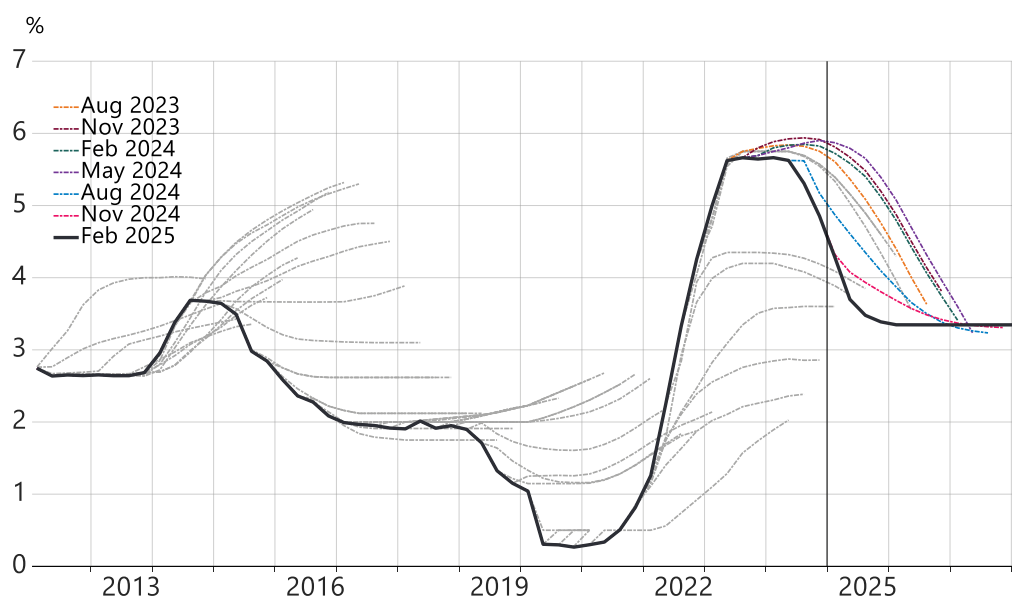
So, you will sometimes hear us talk about a significant change in ‘inflationary pressure’, yet our inflation forecast looks more or less the same as in the previous *Statement*. In this case, we will have shifted our outlook for the OCR to offset the change in inflationary pressure, leaving the outlook for inflation broadly unchanged. For example, if our OCR track is higher, our view of underlying inflationary pressure has increased and *vice versa*.

Figure 1: Reserve Bank annual CPI inflation projections since 2012



Source: Statistics New Zealand and RBNZ.

Figure 2: Reserve Bank OCR projections since 2012



Source: Statistics New Zealand and RBNZ.

The many benefits of publishing our forecasts

There are several reasons why we publish detailed macroeconomic forecasts, including a projection for the OCR, with every *Monetary Policy Statement*. In addition to guiding our internal decision making, our economic forecasts also help improve transparency, accountability and ultimately the effectiveness of monetary policy.

Enhanced transparency and accountability

We have been given operational independence to formulate and implement monetary policy in New Zealand. To preserve our social licence, we need to be transparent in communicating our monetary policy strategy and decisions. Regularly presenting MPC's policy strategy, and the forecasts and judgements that lie behind it, enhance our accountability to the public and government.

In addition, publishing an OCR track that reflects the MPC's policy strategy also enables increased clarity in our communications. We don't need to rely solely on text in our policy documents to describe the policy outlook – instead we can rely on our OCR projection to tell a large part of the policy story.

Our forecasts also help us in looking back and assessing our monetary policy decisions and learning from our experiences. We regularly assess our forecast accuracy and the factors that have driven forecast errors. This provides input into our research programme, which ultimately improves our understanding of the economy and our decision-making process.

Enhanced policy effectiveness

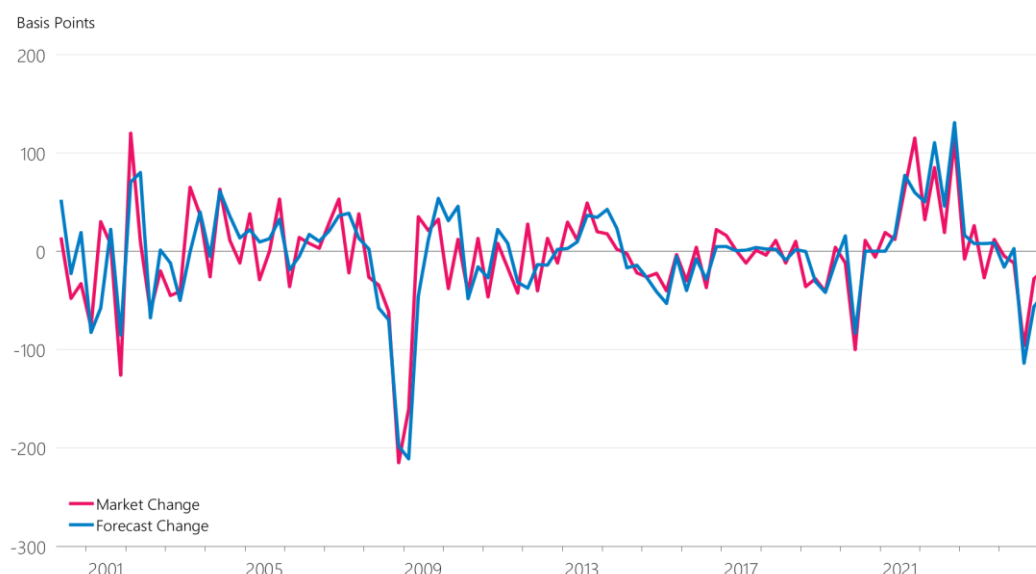
Regularly publishing economic projections helps make it clear to the public that the MPC are conducting monetary policy to achieve price stability in the New Zealand economy. In turn, this helps anchor inflation expectations to the midpoint of the inflation target, which is key in maintaining low and stable inflation over the medium term.

Our forecasts also influence wholesale and retail interest rates in New Zealand, helping to steer interest rates to levels consistent with price stability. Publishing a projection for the future path of the OCR gives financial markets greater clarity over how we plan to meet our commitment to price stability.

Publishing an endogenous OCR track also conveys how the MPC responds to new information over time – in other words, our reaction function. If this is sufficiently clear, then participants in financial market will adjust their OCR expectations, and financial conditions more broadly, as new information becomes available. As a result, financial conditions often adjust consistent with maintaining low and stable inflation even before we publish an updated forecast, which improves monetary policy effectiveness.

One way to assess how well market participants understand our reaction function is by comparing actual changes in our OCR projection with changes in market pricing for the OCR through time. If our reaction function is clear, market changes in the OCR outlook will be consistent with our own. Figure 3 shows that this has generally been the case over the past couple of decades, suggesting that market participants have correctly anticipated how we will respond to new events.

Figure 3: Net change in market and forecast implied 2-year swap rate between MPS



Source: LSEG, RBNZ Estimates

Note: Market change is the change in the 2-year swap rate from the close on the day of the previous MPS to the close on the day before the current MPS. Forecast change is the change in the 2-year swap rate implied by our forecasts from one MPS to another. If markets had a perfect understanding of our reaction function, and the same economic outlook, then market pricing would move perfectly in synch with our forecasts.

Conclusion

To sum up, monetary policy takes time to affect the economy, so we must look ahead to decide how to respond today. It follows that forecasting sits at the heart of how we set monetary policy at Te Pūtea Matua. By embedding forecasting within our flexible inflation targeting framework, we ensure that monetary policy decisions support medium-term price stability and the well-being of New Zealanders.

Because economic data is far from ideal, nowcasting is an important tool. By improving our understanding of the present, nowcasting enhances the accuracy of our broader economic forecasts and helps us make better-informed policy decisions.

While forecasting is challenging and subject to uncertainty, it remains an essential tool for shaping expectations, guiding policy, and maintaining economic stability. By clearly communicating our conditional forecasts, we enhance transparency, accountability and the effectiveness of our policies. It requires the use of models, data analysis, and expert judgment to anticipate future economic conditions.

And lastly, while forecasts are never perfect, they provide a structured way to think about the economy's future path and guide monetary policy decisions. The forecasts are always conditional. If the facts change, we may need to change our minds so that we always get inflation back to target consistent with our purpose of enhancing the wellbeing and prosperity of New Zealanders.